
Polynomial Links as Parameter-Tied Rank Lifts for Tucker Tensors

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Abstract

Tucker decomposition is a compact model for multiway data, but its map from a latent tensor to measurements is linear. Nonlinear measurement responses can therefore make a low-rank latent tensor appear higher-rank in the measured tensor. We propose Nonlinear Tucker Decomposition with Polynomial Links (NTD-PL), a Tucker enhancement that learns a polynomial link jointly with a Tucker backbone. NTD-PL keeps the backbone ranks fixed and adds only scalar link coefficients. Its polynomial powers induce a parameter-tied rank lift: each degree forms Veronese-lifted Tucker factors tied to the same latent tensor, rather than an untied high-rank correction.

We analyze this structure through Veronese rank expansion, separation from matched-rank Tucker, and approximation results for nonlinear responses. We optimize a masked reconstruction and completion objective by alternating Link Refresh, a small ridge update for the scalar link, with Linked-Backbone Descent and degree continuation. Experiments on controlled tensors, hyperspectral reconstruction and completion, rank-lift and Link Refresh diagnostics, and external cross-domain cases show scoped gains: NTD-PL helps when the residual left by matched-rank Tucker is coherent with a nonlinear response.

1 Introduction

Tensor decompositions provide compact models for multiway data in scientific data analysis, signal processing, and masked multiway learning [Kolda and Bader, 2009, Sidiropoulos et al., 2017, Liu et al., 2013, Liu and Moitra, 2020]. Tucker decomposition is a standard choice because it separates a small core from mode-wise factors. It uses a linear map from the Tucker latent tensor to the measured tensor: the reconstructed Tucker value is fitted directly to the measurement. This assumption is natural for many data sets, but it can be mismatched to measurement or data-generation pipelines with response curves, saturation, calibration effects, or material-dependent nonlinearities [Debevec and Malik, 1997, Grossberg and Nayar, 2004, Bioucas-Dias et al., 2012, Heylen et al., 2014, Luo et al., 2024]. In such settings, the latent signal may be well organized by a low-rank Tucker tensor while the measured tensor appears to have larger multilinear rank after a nonlinear response.

A direct remedy is to increase Tucker rank, but this changes the multilinear budget and expands the core and factors throughout the model. More flexible nonlinear tensor decoders based on kernels, Gaussian processes, or neural networks can also be useful, but they address a broader problem and often make the relation to a Tucker backbone less explicit [Xu et al., 2011, Zhe et al., 2016, Liu et al., 2017, Saragadam et al., 2024, Ahn et al., 2024]. We ask a narrower question: can a Tucker model keep its chosen backbone ranks fixed while learning a response that explains coherent nonlinear residuals left by matched-rank Tucker?

We instantiate this idea with a polynomial link. Let

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$$

37 be a Tucker latent tensor. Instead of fitting the measured tensor directly by $\mathcal{S}$, we model

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p},$$

38 where $\odot p$ denotes the Hadamard power. We call the model Nonlinear Tucker Decomposition
 39 with Polynomial Links (NTD-PL). It contains Tucker as the degree-one case, keeps the Tucker
 40 backbone and multilinear ranks fixed, and adds only $P + 1$ scalar link coefficients. The higher powers
 41 $\mathcal{S}^{\odot p}$ induce high-order interactions among the same core and factors, so the added expressivity is
 42 parameter-tied to one learned latent tensor.

43 This viewpoint is central to the paper. NTD-PL is best understood as a Tucker enhancement and
 44 shared-backbone rank lift, not as a generic nonlinear decoder. A degree- p power of the latent tensor
 45 behaves like an ordinary Tucker tensor whose factors are polynomial feature lifts of the original
 46 factors, but those lifted features remain tied to one learned backbone. The new degrees therefore
 47 reshape the low-rank geometry through the polynomial link, rather than adding an untied high-rank
 48 tensor block. This is also why the link is not a post-hoc calibration layer.

49 Our claims are intentionally scoped. Increasing Tucker rank remains the right comparison when the
 50 residual is ordinary multilinear structure, and we do not claim to dominate neural, kernel, or other
 51 nonlinear tensor factorization methods. The intended regime is more specific: the residual left by
 52 matched-rank Tucker should be coherent with a nonlinear response.

53 This scoped regime is also supported empirically. In controlled tensors, the gain grows with the
 54 injected nonlinear residual energy and with the degree needed to approximate the response. On
 55 real hyperspectral tensors, NTD-PL improves matched-rank Tucker in both full reconstruction and
 56 held-out completion. More importantly, a one-step Link Refresh diagnostic predicts when NTD-PL
 57 will help: across HSI scenes and non-HSI tensor units, larger diagnostic reduction is associated with
 58 larger NTD-PL gains, while low-yield units show little or no benefit. These results suggest that
 59 nonlinear responses are identifiable structure in several real tensor families.

60 **Contributions.** We make the following contributions.

- 61 • **Model.** We propose NTD-PL, a Tucker enhancement with a polynomial link on a Tucker latent ten-
 62 sor. It keeps the Tucker backbone and multilinear ranks fixed while adding only scalar coefficients
 63 for a parameter-tied response.
- 64 • **Theory.** We characterize the induced model class through Veronese rank expansion, separation
 65 from matched-rank Tucker, and approximation results for nonlinear responses.
- 66 • **Optimization.** We derive a masked objective for reconstruction and completion, and optimize it
 67 with Link Refresh, Linked-Backbone Descent, and degree continuation.
- 68 • **Experiments.** We evaluate controlled nonlinear residuals, hyperspectral reconstruction and com-
 69 pletion, rank-lift and Link Refresh diagnostics, and external cross-domain boundary cases.

70 2 Related Work

71 **Low-rank tensor models and Tucker backbones.** CP, Tucker, tensor train, and tensor ring
 72 decompositions provide compact representations for multiway data through different rank notions
 73 and factor topologies [Kolda and Bader, 2009, Sidiropoulos et al., 2017, De Lathauwer et al., 2000,
 74 Oseledets, 2011, Zhao et al., 2016]. Recent completion work has improved scalability, sampling
 75 guarantees, and visual-data priors while still mostly changing the tensor rank model, tensor-network
 76 topology, or low-rank regularizer [Liu and Moitra, 2020, Wang et al., 2021, Zheng et al., 2021].
 77 Tucker is the closest baseline for NTD-PL because it uses the same core-factor multilinear backbone.
 78 NTD-PL does not primarily compete by changing the rank tuple, replacing the factor topology, or
 79 moving to another tensor-network format. Instead, it keeps the Tucker backbone and multilinear
 80 ranks fixed and changes the measurement map on top of the Tucker latent tensor.

81 **Nonlinear tensor factorization and neural/kernel decoders.** Nonlinear tensor models include
 82 generalized likelihood and link formulations, Bayesian and kernelized tensor factorizations, Gaussian-
 83 process tensor models, and neural tensor decoders [Hong et al., 2020, Schein et al., 2016, Xu et al.,

2011, Zhe et al., 2016, Liu et al., 2017, Pan et al., 2020, Saragadam et al., 2024, Varolgunes et al., 2023, Ahn et al., 2024, Luo et al., 2024]. Related imaging methods also learn linear or nonlinear transforms under low-rank tensor priors [Luo et al., 2022, Wang et al., 2024, Xu et al., 2024]. These methods can be more flexible than a scalar polynomial link, but this flexibility often comes with heavier inference, additional hyperparameters, or a less transparent finite-dimensional relation to Tucker. NTD-PL is complementary: it deliberately uses a low-dimensional scalar response rather than a general kernel, Gaussian-process, neural decoder, or learned transform, so that the induced effect can be analyzed relative to a fixed Tucker backbone.

Polynomial feature lifts and high-order interactions. Polynomial and Volterra-style expansions are classical tools for representing nonlinear responses [Volterra, 1959, Boyd and Chua, 1985]. Related matrix and tensor methods use polynomial feature maps, quadratic factorizations, or explicit high-order interactions to enlarge a linear model [Zhai et al., 2024, Ruegamer, 2024, Liu et al., 2015, Deng et al., 2016]. Such constructions can make nonlinear interactions explicit, but untied high-order parameterizations can increase parameter dimension and contraction cost. NTD-PL does not introduce an independent high-order tensor parameter. Its polynomial powers are powers of one learned Tucker latent tensor, inducing parameter-tied high-order interactions and Veronese-style backbone-tied rank lifts.

Positioning. Existing nonlinear tensor models either use broad nonlinear decoders, modify likelihoods or kernels, or explicitly parameterize high-order interactions. What is less developed is a Tucker-preserving, low-dimensional, analyzable polynomial link that yields a parameter-tied rank lift and supports masked learning. NTD-PL trades general nonlinear flexibility for structure, interpretability, and a direct finite-dimensional rank-lift view. Its expected benefit is therefore scoped to settings where matched-rank Tucker leaves a coherent nonlinear residual explained by a scalar response.

3 Model

3.1 Model and Measurement Assumption

Let $\mathcal{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ be a measured tensor and let

$$\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket, \quad \mathcal{X} \in \mathbb{R}^{R_1 \times \dots \times R_N}, \quad \mathbf{A}^{(n)} \in \mathbb{R}^{I_n \times R_n}. \quad (1)$$

Our measurement assumption is that the measured tensor is generated from this low-rank Tucker latent tensor through a scalar response, up to noise. NTD-PL models this response by a polynomial:

$$Y_{\mathbf{i}} = f_{\beta}(S_{\mathbf{i}}) + \varepsilon_{\mathbf{i}}, \quad f_{\beta}(s) = \sum_{p=0}^P \beta_p s^p, \quad \mathbf{i} = (i_1, \dots, i_N). \quad (2)$$

Equivalently, the tensor prediction is

$$\hat{\mathcal{Y}} = f_{\beta}(\mathcal{S}) = \sum_{p=0}^P \beta_p \mathcal{S}^{\odot p}, \quad \beta \in \mathbb{R}^{P+1}. \quad (3)$$

The constant term β_0 captures affine response shifts, the linear term $\beta_1 \mathcal{S}$ is the usual Tucker channel, and terms with $p \geq 2$ activate nonlinear channels on the same latent tensor. Standard Tucker is recovered by setting $\beta_1 = 1$ and $\beta_p = 0$ for $p \neq 1$. Thus NTD-PL changes the measurement map while leaving the Tucker backbone and its multilinear ranks fixed.

3.2 Backbone-Tied High-Order Interactions

Although f_{β} is scalar, the term $\mathcal{S}^{\odot p}$ is not merely an independent output calibration. We arrive at it by first defining an explicit p -fold Tucker extension: instead of mapping one latent index per mode, the mode- n projection is a higher-order tensor $\mathcal{A}^{(n,p)} \in \mathbb{R}^{I_n \times R_n^p}$ that couples p latent indices. The case $p = 1$ is ordinary Tucker, while $p = 2$ gives a quadratic Tucker-style interaction between two latent Tucker tuples. Appendix A.1 gives the formal construction and tensor-network view.

The explicit construction is a reference model, not the parameterization we train. Its projection tensors grow as $I_n R_n^p$, and each degree would add a new set of mode projections. NTD-PL keeps the

125 same high-order interaction pattern by tying the p -fold projection to the original Tucker factor:

$$\mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)} = \prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)}.$$

126 Under this tying, the degree- p channel is exactly S_i^p . Thus it still couples p latent Tucker index tuples,
 127 but all copies share the same mode factors. Across degrees, the only new free parameters are the
 128 scalar coefficients $\{\beta_p\}$, which mix these tied channels.

129 The latent tensor $\mathcal{S}$ and the coefficients β are fitted jointly, so the low-rank backbone is optimized
 130 together with the polynomial channels. Section 4 analyzes the induced rank expansion and the
 131 resulting separation from standard Tucker.

132 3.3 Masked Least-Squares Objective

133 Let $\Omega \in \{0, 1\}^{I_1 \times \dots \times I_N}$ denote the observed-entry mask and let $|\Omega| = \sum_i \Omega_i$. We estimate the core,
 134 factors, and link coefficients by the regularized masked objective

$$\min_{\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta} \frac{1}{2|\Omega|} \|\Omega \odot (\mathcal{Y} - f_{\beta}(\mathcal{S}))\|_F^2 + \frac{\lambda_X}{2} \|\mathcal{X}\|_F^2 + \sum_{n=1}^N \frac{\lambda_n}{2} \|\mathbf{A}^{(n)}\|_F^2 + \frac{\lambda_{\beta}}{2} \|\beta\|_2^2. \quad (4)$$

135 The same formulation covers full reconstruction by taking $\Omega \equiv \mathbf{1}$ and tensor completion by restricting
 136 the loss to the observed entries. The quadratic penalties regularize the Tucker backbone and the
 137 low-dimensional polynomial link.

138 4 Theory

139 The theory addresses the rank-lift view raised by the model definition. A scalar link has few free
 140 parameters, but its powers can induce ordinary Tucker tensors with larger multilinear ranks. We
 141 first make this induced structure explicit, then give a separation from matched-rank Tucker and an
 142 approximation route for nonlinear responses.

143 4.1 Rank Expansion and Separation

144 **Proposition 1** (Shared-backbone rank expansion). *Let $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$ have Tucker rank*
 145 *at most $(R_1, \dots, R_N)$. For every integer $p \geq 1$, $\mathcal{S}^{\odot p}$ admits a Tucker representation with mode ranks*
 146 *at most*

$$\left(\binom{R_1 + p - 1}{p}, \dots, \binom{R_N + p - 1}{p} \right). \quad (5)$$

147 *More explicitly, the ordinary Tucker factors of $\mathcal{S}^{\odot p}$ can be chosen as Veronese lifts of the original*
 148 *factors. For $\mathbf{A} \in \mathbb{R}^{I \times R}$, define $\mathbf{A}^{(p)} \in \mathbb{R}^{I \times D}$, $D = \binom{R+p-1}{p}$, by*

$$A_{i,\alpha}^{(p)} = \prod_{r=1}^R A_{i,r}^{\alpha_r}, \quad (6)$$

149 *where $\alpha \in \mathbb{N}^R$ ranges over $|\alpha| = p$. Then $\mathcal{S}^{\odot p}$ factors through $\mathbf{A}^{(1)(p)}, \dots, \mathbf{A}^{(N)(p)}$.*

150 This proposition makes the tying in Section 3.2 explicit in standard Tucker language: a degree- p
 151 channel is an ordinary Tucker tensor whose mode factors are polynomial feature lifts of the same
 152 backbone factors. The link therefore changes rank structure through tied features, not through an
 153 independent high-order parameter block.

154 **Theorem 1** (Tensorial separation from standard Tucker). *Let $N \geq 3$, $R \geq 2$, $p \geq 2$, and suppose*
 155 *$I_n \geq D$ for every n , where $D = \binom{R+p-1}{p}$. Then there exists an order- N tensor $\mathcal{Y} \in \mathbb{R}^{I_1 \times \dots \times I_N}$ that*
 156 *is representable by a degree- p NTD-PL model with Tucker backbone rank $(R, \dots, R)$, but whose*
 157 *exact standard Tucker rank is*

$$(D, \dots, D). \quad (7)$$

158 *Equivalently, a degree- p polynomial link can generate tensors whose multilinear ranks are*
 159 *$(\binom{R+p-1}{p}, \dots, \binom{R+p-1}{p})$ from the same Tucker backbone of rank $(R, \dots, R)$.*

The proof in Appendix B uses a CP-diagonal Tucker backbone and generic full-rank Veronese feature matrices. The theorem is an existence separation, not a universality claim over all high-rank Tucker tensors: it identifies a structured high-rank subset generated by a low-rank backbone.

4.2 Approximation Under Nonlinear Measurement Responses

Theorem 2 (Polynomial generation). *Suppose $\mathcal{Y} = g(\mathcal{S}^*)$, where $\mathcal{S}^*$ has Tucker rank at most $(R_1, \dots, R_N)$ and g is a polynomial of degree at most P . Then degree- P NTD-PL with the same Tucker rank can represent $\mathcal{Y}$ exactly.*

Theorem 3 (Analytic link approximation). *Suppose $\mathcal{Y} = g(\mathcal{S}^*)$, where $\mathcal{S}^*$ has Tucker rank at most $(R_1, \dots, R_N)$ and $\|\mathcal{S}^*\|_\infty \leq B$. If g is analytic on the complex disk $D(0, \rho)$ with $B < \rho$, and $M_\rho = \sup_{|z|=\rho} |g(z)|$, then there exists a degree- P NTD-PL model with the same Tucker backbone satisfying*

$$\frac{1}{\sqrt{M}} \|\mathcal{Y} - \hat{\mathcal{Y}}\|_F \leq M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}, \quad M = \prod_{n=1}^N I_n. \quad (8)$$

Thus the same finite link that induces a tied rank lift also gives a direct approximation route for smooth responses on a bounded latent range. The statement uses the simple Taylor truncation bound; sharper polynomial approximation bounds can replace it under stronger assumptions.

Together, the results clarify the scope of NTD-PL. Polynomial links can approximate nonlinear responses while keeping the same latent Tucker rank, and their powers induce Veronese-lifted Tucker factors. This explains how a scalar link can change multilinear rank structure without introducing untied high-order tensor parameters.

5 Optimization

We optimize the masked objective in Section 3.3 by alternating between the scalar link and the Tucker backbone. The nonlinear part is low-dimensional once the latent Tucker tensor is fixed, while the backbone update remains a standard differentiable Tucker fitting step through the current link. We call the two stages *Link Refresh* and *Linked-Backbone Descent*.

Link Refresh. Fix the current latent tensor $\mathcal{S} = [\mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)}]$. For active degree $Q \leq P$, updating the polynomial coefficients reduces to a small masked ridge regression,

$$\min_{\beta_{0:Q}} \frac{1}{2|\Omega|} \|\mathcal{Y}_\Omega - \Phi_{\Omega,Q}(\mathcal{S})\beta_{0:Q}\|_2^2 + \frac{\lambda_\beta}{2} \|\beta_{0:Q}\|_2^2, \quad (9)$$

where $\Phi_{\Omega,Q}(\mathcal{S})$ contains the observed powers $[1, s, \dots, s^Q]$. This step is cheap because it solves only a $(Q+1) \times (Q+1)$ linear system. It also gives an interpretable view of the link: once the backbone exposes scalar latent values over the tensor, the refresh estimates the best scalar response on the observed set.

Linked-Backbone Descent. With β fixed, the Tucker parameters are updated through the current nonlinear measurement map. The masked residual is first propagated to the latent tensor as

$$T = \frac{1}{|\Omega|} \Omega \odot (f_\beta(\mathcal{S}) - Y) \odot f'_\beta(\mathcal{S}), \quad (10)$$

and then backpropagated through the Tucker reconstruction to update $\mathcal{X}$ and $\{\mathbf{A}^{(n)}\}$. In implementation we use Adam for these first-order steps, but the update itself is just standard Tucker reverse contractions composed with the current link. The backbone is therefore not calibrated after fitting; it is jointly reshaped under the nonlinear link.

Stabilization. Directly activating all polynomial degrees from the start is brittle, because higher powers of $\mathcal{S}$ amplify scale ambiguity. We therefore begin from the Tucker-compatible degree-one model and activate higher degrees gradually. When a new degree is switched on, its coefficient is initialized at zero, so the model expands from the current lower-degree solution without changing the prediction abruptly. We also normalize Tucker factor scales and absorb the corresponding

scales into the core, which fixes a Tucker gauge without changing S . For the coefficient solve, the implementation may regress on a centered and scaled latent variable and then map the solution back to the original power basis. Appendix Figure 6 shows representative training curves where activating the relevant higher degrees produces systematic loss drops. Full gradients, stabilization details, costs analysis and pseudocode are given in Appendix A.2.

Masked uniform convergence. The masked objective also admits a standard sampling interpretation. Let $m = |\Omega|$ and let $\Theta \subset \mathbb{R}^d$ be a bounded NTD-PL parameter set, where $d = \prod_{n=1}^N R_n + \sum_{n=1}^N I_n R_n + (P + 1)$. If the per-sample losses are bounded by $0 \leq \ell_i(\theta) \leq B_\ell$ and G -Lipschitz in θ , then with probability at least $1 - \delta$,

$$\sup_{\theta \in \Theta} |\mathcal{L}(\theta) - \hat{\mathcal{L}}_m(\theta)| \leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}, \quad (11)$$

for every $0 < \varepsilon \leq B_\Theta$. This is a standard uniform convergence bound, not a sharp minimax result, but it clarifies that masked training controls the full-tensor risk with a complexity term determined by the tied NTD-PL parameter count. The proof is given in Appendix B.4.

6 Experiments

The experiments address three mechanism questions. First, under controlled nonlinear residuals, does NTD-PL behave as intended: remaining close to Tucker in the linear regime while gaining when the residual is link-like and the active polynomial degree is expressive enough? Second, on real hyperspectral tensors, does the polynomial link provide a compact fixed-backbone enhancement that improves both representation and held-out completion, rather than merely spending more parameters? Third, can a frozen-backbone Link Refresh diagnostic identify, before full NTD-PL training, which tensor units contain residual structure that a scalar link can exploit? Unless stated otherwise, real tensors are globally max-normalized and used without spatial downsampling or cropping. We report $RMSE$ and SAM for reconstruction, and $RMSE^*$ and SAM^* only on held-out entries for completion. NTD-PL is compared mainly against same-rank Tucker, except in the rank-lift ablation.

6.1 Controlled Nonlinear Residuals

These synthetic experiments are mechanism checks rather than generic benchmarks. We generate measured tensors as

$$Y_\alpha = \sqrt{1 - \alpha} S^* + \sqrt{\alpha} \tilde{R}^*,$$

where S^* is a low-rank Tucker tensor and $\tilde{R}^*$ is the normalized component of a scalar nonlinear response orthogonal to S^* . Thus α controls nonlinear residual energy, not just the raw amplitude of a nonlinearity. Figure 1 shows the expected transition: NTD-PL is Tucker-like when the residual is nearly linear and separates as nonlinear residual energy increases. Tucker can lag behind CP, but NTD-PL then pulls ahead systematically. Figure 2 shows the complementary degree effect: gains emerge when the active polynomial degree can match or approximate the response complexity. Degree continuation and coefficient-contribution details are deferred to Appendix C.1, where Table 4 shows the linear consistency negative control.

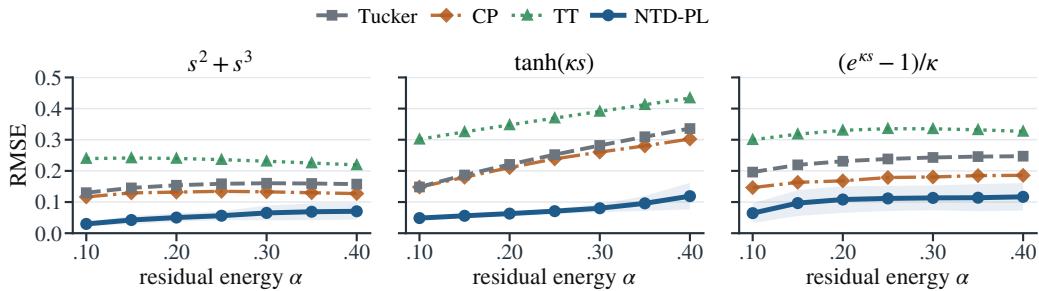


Figure 1: Controlled nonlinear residual sweep. NTD-PL stays Tucker-like near the linear regime and separates as α , the nonlinear residual energy, increases. Lower RMSE is better.

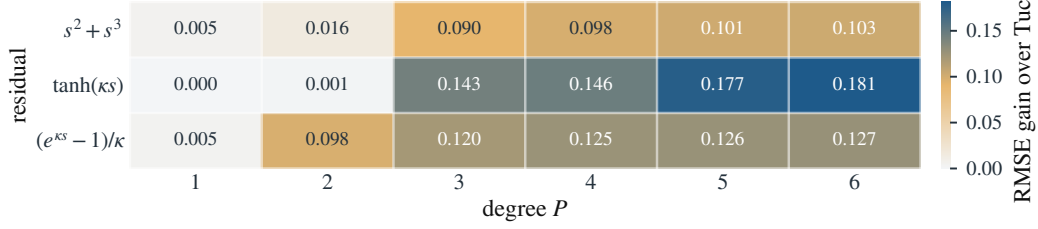


Figure 2: Degree-dependent gains at $\alpha = 0.25$. Cell values are Tucker minus NTD-PL RMSE; gains appear when the active degree is sufficient for the response complexity.

234 Taken together, these controlled experiments answer the first question. NTD-PL does not obtain gains
 235 simply by adding a polynomial link: it stays close to Tucker when the source is linear or nearly linear,
 236 and its advantage appears when the residual is both nonlinear and expressible by the active degrees.
 237 The synthetic results therefore isolate the intended mechanism before moving to real tensors.

238 6.2 Tucker Enhancement on Real Hyperspectral Tensors

239 This section tests the same fixed-rank enhancement on real hyperspectral tensors, in both full
 240 reconstruction and held-out completion. We use CAVE [Columbia CAVE Laboratory, n.d., Yasuma
 241 et al., 2010], whose multiple real scenes provide a compact spectral-spatial mechanism check.

242 **Representation enhancement.** Full reconstruction checks representation capacity rather than
 243 generalization. Figure 3 shows that, at matched Tucker backbone rank and similar parameter budget,
 244 NTD-PL reduces RMSE and SAM. The gap shrinks as Tucker rank grows, while Table 1 shows that
 245 Tucker needs substantially more rank and parameters to match NTD-PL RMSE. NTD-PL also usually
 246 keeps a clearer SAM advantage.

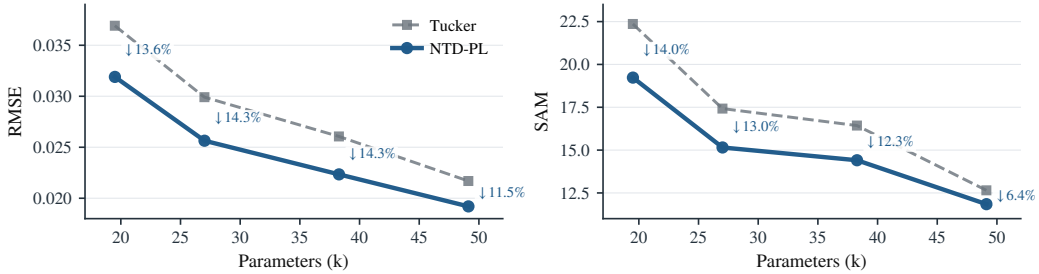


Figure 3: CAVE full reconstruction. At matched backbone rank, NTD-PL reduces RMSE and SAM; annotations show paired gains over Tucker. The gap narrows with Tucker rank, consistent with fixed-rank enhancement. Points average 15 scenes.

Table 1: Rank-lifted Tucker comparison on CAVE reconstruction. For each NTD-PL backbone rank, we report the first higher-rank Tucker model matching NTD-PL RMSE and its extra parameters.

NTD-PL reference				First Tucker matching RMSE				Extra params
Rank	Params	RMSE↓	SAM↓	Rank	Params	RMSE↓	SAM↓	
(18, 18, 3)	20k	0.0319	19.23	(28, 28, 3)	31k	0.0316	21.41	+59.5%
(24, 24, 4)	27k	0.0256	15.15	(35, 35, 4)	41k	0.0254	16.26	+51.3%
(33, 33, 4)	38k	0.0223	14.41	(49, 49, 4)	60k	0.0222	15.28	+56.5%

247 **Held-out enhancement.** Held-out completion tests transfer to unseen entries. Table 2 reports
 248 random-mask CAVE completion, with metrics computed only on missing entries. At the same
 249 rank budget, NTD-PL gives consistent RMSE* gains and usually improves SAM*; paired sign and
 250 Wilcoxon tests in Appendix C.2 show the same scene-level pattern.

251 These CAVE results answer the second question. The polynomial link improves matched-rank Tucker
 252 not only as a representation-capacity check, but also on held-out entries under random missingness.
 253 The rank-lift comparison further shows that the gain is not merely a parameter-count effect: at the

Table 2: Held-out CAVE completion at rank (33, 33, 4) under random masks. Metrics use missing entries only; gains are paired improvements over matched-rank Tucker.

Missing rate	RMSE*↓			SAM*↓		
	Tucker	NTD-PL	Gain	Tucker	NTD-PL	Gain
0.1	0.0263 ± 0.0179	0.0229 ± 0.0158	12.9%	11.82 ± 5.33	8.36 ± 3.38	29.3%
0.3	0.0263 ± 0.0179	0.0230 ± 0.0158	12.5%	14.75 ± 6.33	12.47 ± 5.06	15.5%
0.5	0.0264 ± 0.0179	0.0230 ± 0.0158	12.9%	15.56 ± 6.85	13.48 ± 5.75	13.4%
0.7	0.0266 ± 0.0180	0.0219 ± 0.0128	17.7%	16.02 ± 7.21	13.79 ± 5.43	13.9%

same Tucker backbone rank, the tied link supplies a compact nonlinear enhancement that higher-rank Tucker can only match with substantially more ordinary multilinear capacity.

6.3 Diagnosing Nonlinear Low-Rank Structure Across Data

NTD-PL is meant to be diagnosable rather than uniformly beneficial: it should help when matched-rank Tucker leaves residual structure that a scalar link can reduce. We use D_{link} as a cheap frozen-backbone probe: fit Tucker, apply one Link Refresh step from Section 5, and record how much Tucker residual energy this one-step link removes on the evaluation entries. Appendix C.3 gives the exact formula.

For CAVE completion, Table 3 groups scenes by D_{link} tercile. Higher scores correspond to larger held-out gains across missing rates, with Spearman correlations around 0.829–0.889; Appendix Figure 7 gives scene-level scatters.

Table 3: CAVE completion diagnostic. Scenes are grouped by D_{link} tercile; entries are mean NTD-PL RMSE* gains over matched-rank Tucker.

Missing rate	Low D_{link}	Mid D_{link}	High D_{link}	Spearman ρ_s
0.1	0.75%	7.92%	12.63%	0.829
0.3	-1.13%	9.10%	13.12%	0.871
0.5	-1.09%	9.16%	13.18%	0.889
0.7	-0.62%	9.22%	13.07%	0.868

We apply the same protocol beyond HSI. Figure 4 plots D_{link} against RMSE gain. Across all three families, larger D_{link} consistently identifies units with larger NTD-PL gains; object-view tensors show the strongest separation, action video follows the same trend, and natural images retain a positive but lower-yield regime. Appendix Table 7 gives the numeric statistics behind Figure 4.

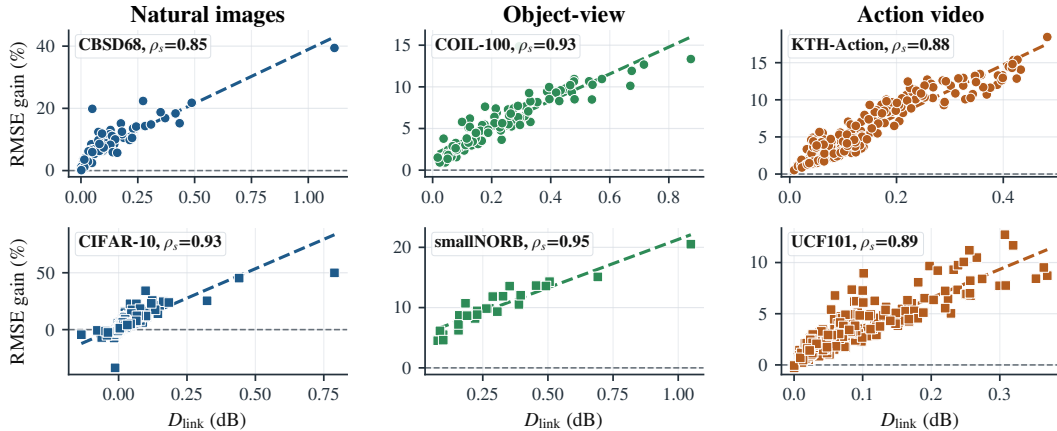


Figure 4: Per-unit cross-domain diagnostic correlation. Each panel plots D_{link} against NTD-PL RMSE gain over independently fitted tensor units.

These diagnostic results answer the third question. A single frozen-backbone Link Refresh score is already predictive of where full NTD-PL will help, across both CAVE completion and non-HSI tensor units. This supports the scoped claim of the paper: NTD-PL is useful when matched-rank

272 Tucker leaves residual structure that is coherent with a shared scalar response, and low diagnostic
273 yield correctly signals cases where the polynomial link should provide little benefit.

274 7 Limitations

275 NTD-PL assumes that a common nonlinear response explains a coherent part of the residual left by
276 matched-rank Tucker. A single scalar link may be less effective when nonlinearities vary strongly
277 across spatial regions, spectral bands, modes, or samples; when the remaining error is ordinary
278 multilinear structure rather than a scalar response, increasing Tucker rank remains the more direct
279 modeling choice.

280 Polynomial degree adds approximation capacity but can amplify Tucker scale ambiguity and con-
281 ditioning issues in the power basis, motivating the ridge-regularized Link Refresh step, factor
282 normalization, and degree continuation. The joint problem also remains nonconvex: Link Refresh is
283 a small ridge regression with fixed latent tensor, but the backbone and link are learned jointly, and we
284 do not claim global optimality.

285 Our strongest empirical evidence is from controlled nonlinear residuals and hyperspectral reconstruc-
286 tion and completion. Broader validation on large sparse tensors, temporal or multimodal tensors, and
287 domain-varying nonlinear responses is left for future work.

288 8 Conclusion

289 We asked whether a Tucker model can keep its chosen backbone ranks fixed while learning a
290 structured response that explains coherent nonlinear residuals left by matched-rank Tucker. NTD-PL
291 gives a positive but scoped answer. It places a polynomial link on a shared Tucker latent tensor,
292 so each higher-degree channel is not an independent high-rank correction but a Veronese-lifted,
293 parameter-tied rank lift of the same backbone. In this way, the model preserves the core-factor
294 structure and rank budget of Tucker while adding only a small number of scalar link coefficients.

295 The analysis and optimizer are built around this shared-backbone view. The theory shows how
296 polynomial powers induce tied rank expansion, separates the resulting model class from matched-
297 rank Tucker, and provides approximation guarantees for nonlinear measurement responses. The
298 optimization procedure alternates a small Link Refresh ridge solve with Linked-Backbone Descent
299 and degree continuation, making the link a jointly learned part of the representation rather than a
300 post-hoc calibration layer.

301 The experiments identify the regime in which this additional structure is useful. Controlled residual
302 experiments show that NTD-PL stays close to Tucker in linear or nearly linear settings and gains
303 when the residual is nonlinear and expressible by the active polynomial degrees. CAVE reconstruction
304 and completion show that the same fixed-backbone enhancement improves both representation and
305 held-out prediction, while the rank-lift ablation shows that the gain is not merely a parameter-count
306 effect. Finally, the Link Refresh diagnostic predicts where full NTD-PL helps across HSI and non-HSI
307 tensor units. These results position NTD-PL not as a universal nonlinear tensor decoder, but as a
308 compact Tucker enhancement for tensors whose matched-rank residuals are coherent with a shared
309 scalar response. Future work may extend this idea to adaptive links, mode- or region-dependent
310 responses, and scalable sparse implementations.

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428 A Additional Model Details

429 A.1 Explicit p -Fold Tucker Motivation

430 This section expands the motivation behind the explicit p -fold Tucker channel in Section 3.2. The
431 construction is best viewed as the tensor analogue of adding polynomial interactions to a linear map.
432 In the matrix case, one may move from a linear model $\mathbf{y} = \mathbf{A}\mathbf{x}$ to a quadratic model

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathcal{A}(\mathbf{x}, \mathbf{x}),$$

433 where the second term couples pairs of latent coordinates. The corresponding question for Tucker is:
434 if standard Tucker is the multilinear low-rank map for multiway data, what is the direct p -th order
435 interaction analogue?

436 For a Tucker backbone, an explicit p -fold interaction can be written as

$$\tilde{Y}_i^{(p)} = \sum_{\mathbf{r}^{(1)}, \dots, \mathbf{r}^{(p)}} \left(\prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \right) \prod_{n=1}^N \mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)}. \quad (12)$$

437 Here each mode projection depends on p latent indices instead of one. For $p = 1$, this reduces to the
438 ordinary Tucker entry formula. For $p = 2$, the same core appears twice and the mode projections
439 couple two latent tuples, giving a quadratic Tucker-style channel. Figure 5 illustrates this progression.

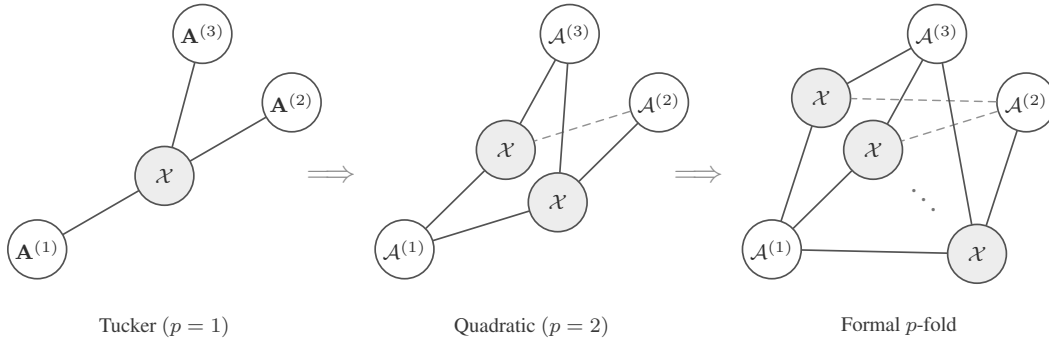


Figure 5: Tensor-network view of a formal p -fold Tucker extension. The construction keeps the Tucker idea of mode-wise generation, but lets each mode projection couple multiple latent index tuples.

440 The explicit construction is useful as a reference model, but it is not the parameterization used by
441 NTD-PL. Its mode projections $\mathcal{A}^{(n,p)}$ have size $I_n R_n^p$, so a direct implementation introduces new

high-order projection tensors and rapidly increases storage and contraction cost. NTD-PL keeps the same high-order interaction pattern by tying the projection tensor to the original Tucker factor:

$$\mathcal{A}_{i_n, r_n^{(1)}, \dots, r_n^{(p)}}^{(n,p)} = \prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)}. \quad (13)$$

Substituting (13) into (12) gives exactly $\mathcal{S}^{\odot p}$, where $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$. Thus the degree- p NTD-PL channel is a shared-projection version of the explicit p -fold Tucker prototype.

This is the modeling role of the p -fold Tucker prototype: it defines the high-order Tucker interaction that NTD-PL compresses. The final model does not train the untied tensors $\mathcal{A}^{(n,p)}$; it keeps a single Tucker backbone and adds only the polynomial coefficients $\{\beta_p\}$.

A.2 Optimization Details

This section gives the update equations behind the solver summarized in Section 5. The implementation follows the same structure: one Tucker reconstruction per iteration, a fused evaluation of the polynomial and its derivative, first-order updates of the backbone, and periodic ridge-regression updates of the link coefficients.

Backbone gradients. For a fixed β , define

$$f'_\beta(\mathcal{S}) = \sum_{p=1}^P p \beta_p \mathcal{S}^{\odot(p-1)}. \quad (14)$$

The gradient of the normalized masked data term with respect to the latent tensor $\mathcal{S}$ is

$$\mathbf{T} = \frac{1}{|\Omega|} \Omega \odot (f_\beta(\mathcal{S}) - \mathcal{Y}) \odot f'_\beta(\mathcal{S}). \quad (15)$$

Backpropagating $\mathbf{T}$ through the Tucker reconstruction gives the core gradient

$$\nabla_{\mathcal{X}} \mathcal{L} = \mathbf{T} \times_1 \mathbf{A}^{(1)\top} \times_2 \dots \times_N \mathbf{A}^{(N)\top} + \lambda_{\mathcal{X}} \mathcal{X}. \quad (16)$$

For the n -th factor, let

$$\mathbf{Z}^{(n)} = \left[\mathcal{X} \times_1 \mathbf{A}^{(1)} \dots \times_{n-1} \mathbf{A}^{(n-1)} \times_{n+1} \mathbf{A}^{(n+1)} \dots \times_N \mathbf{A}^{(N)} \right]_{(n)}. \quad (17)$$

Then

$$\nabla_{\mathbf{A}^{(n)}} \mathcal{L} = \mathbf{T}_{(n)} \mathbf{Z}^{(n)\top} + \lambda_n \mathbf{A}^{(n)}. \quad (18)$$

The experiments use Adam for these first-order steps. This choice is not part of the model definition; any comparable first-order optimizer can use the same gradients.

Link update. With $\mathcal{S}$ fixed and active degree $Q = P_{\text{act}}$, the coefficient subproblem is the ridge regression in (9). Its closed-form solution is

$$\beta_{0:Q}^* = (\Phi_{\Omega,Q}^\top \Phi_{\Omega,Q} + |\Omega| \lambda_\beta \mathbf{I})^{-1} \Phi_{\Omega,Q}^\top \mathbf{y}_\Omega. \quad (19)$$

Equivalently, one may absorb $|\Omega|$ into the ridge parameter and solve the usual unnormalized normal equations. If the constant term is disabled, the first column is removed and β_0 is fixed at zero. In the code, this small system is solved either from an explicit Vandermonde design or from moment equations built from cached powers.

Stabilization and continuation. High powers of $\mathcal{S}$ can be sensitive to scale. The implementation therefore normalizes factor columns after backbone updates and absorbs the corresponding scales into the core; this is a gauge fixing of the Tucker parameterization and does not change $\mathcal{S}$. For the coefficient solve, it may regress on the centered and scaled latent variable $(S - \mu)/\sigma$, then convert the solution back to the original power basis of $\mathcal{S}$.

The active degree follows the schedule

$$t_k = \text{round} \left(\frac{kT}{P} \right), \quad k = 1, \dots, P-1.$$

When degree $p+1$ is activated, the previous coefficients and Tucker factors are kept and the new coefficient β_{p+1} is initialized at zero.

Cost. The dominant cost is forming the Tucker reconstruction and its reverse contractions on the observed tensor. For dense tensors this is comparable to standard Tucker optimization at the same rank, plus $O(P|\Omega|)$ work to evaluate the scalar polynomial and its derivative. The link update forms a Vandermonde design or the corresponding moment equations and solves a $(P+1) \times (P+1)$ linear system, with direct cost $O(P^2|\Omega| + P^3)$. Since P is small, this cost is negligible compared with the Tucker contractions.

Algorithm 1 gives the full procedural summary.

Algorithm 1 Masked NTD-PL optimization

Require: observed tensor $\mathcal{Y}$, mask Ω , rank $(R_1, \dots, R_N)$, degree P

```

1: initialize  $\mathcal{X}, \{\mathbf{A}^{(n)}\}$  from Tucker on the masked tensor
2: initialize active degree  $P_{\text{act}} \leftarrow 1$  and  $\beta = (0, 1)$ 
3: for  $t = 1, \dots, T$  do
4:   form  $\mathcal{S} = \llbracket \mathcal{X}; \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(N)} \rrbracket$ 
5:   if a link update is scheduled then
6:     update active coefficients  $\beta_{0:P_{\text{act}}}$  by ridge regression on  $\{(S_i, Y_i) : \Omega_i = 1\}$ 
7:   end if
8:   compute  $f_\beta(\mathcal{S})$  and  $f'_\beta(\mathcal{S})$ 
9:   take first-order steps on  $\mathcal{X}$  and  $\{\mathbf{A}^{(n)}\}$  using the masked gradient
10:  normalize factor scales and absorb them into the core
11:  increase  $P_{\text{act}}$  according to the continuation schedule, initializing the new coefficient at zero
12: end for
13: return  $\mathcal{X}, \{\mathbf{A}^{(n)}\}, \beta$ 

```

B Additional Proofs

B.1 Polynomial and Analytic Approximation

Proof of Theorem 2. Since g has degree at most P , write

$$g(s) = \sum_{p=0}^P c_p s^p. \quad (20)$$

Take the NTD-PL latent tensor to be the true latent tensor, $\mathcal{S} = \mathcal{S}^*$, and set $\beta_p = c_p$. Then, entrywise,

$$\hat{\mathcal{Y}}_{i_1, \dots, i_N} = \sum_{p=0}^P \beta_p (\mathcal{S}_{i_1, \dots, i_N}^*)^p = g(\mathcal{S}_{i_1, \dots, i_N}^*) = \mathcal{Y}_{i_1, \dots, i_N}. \quad (21)$$

Thus $\hat{\mathcal{Y}} = \mathcal{Y}$. □

Lemma 1 (Scalar-to-tensor approximation). *Let $\|\mathcal{S}^*\|_\infty \leq B$. For any scalar function g and any degree- P polynomial q_P ,*

$$\|g(\mathcal{S}^*) - q_P(\mathcal{S}^*)\|_F \leq \sqrt{\prod_{n=1}^N I_n} \sup_{|s| \leq B} |g(s) - q_P(s)|. \quad (22)$$

Proof. Every entry of $\mathcal{S}^*$ lies in $[-B, B]$, so every entrywise approximation error is bounded by $\sup_{|s| \leq B} |g(s) - q_P(s)|$. Squaring, summing over all entries, and taking the square root gives the claim. □

Proof of Theorem 3. Let $g(s) = \sum_{k=0}^\infty a_k s^k$ be the Taylor expansion of g around zero, and choose $q_P(s) = \sum_{k=0}^P a_k s^k$. By Cauchy's estimate,

$$|a_k| \leq \frac{M_\rho}{\rho^k}. \quad (23)$$

494 For $|s| \leq B < \rho$,

$$|g(s) - q_P(s)| \leq \sum_{k=P+1}^{\infty} |a_k| |s|^k \quad (24)$$

$$\leq M_\rho \sum_{k=P+1}^{\infty} (B/\rho)^k = M_\rho \frac{(B/\rho)^{P+1}}{1 - B/\rho}. \quad (25)$$

495 Applying Lemma 1 and taking $\mathcal{S} = \mathcal{S}^*$, $\beta_k = a_k$ for $0 \leq k \leq P$, yields the stated unnormalized
 496 bound. Dividing by $\sqrt{M}$, $M = \prod_{n=1}^N I_n$, gives the normalized form used in Theorem 3. $\square$

497 B.2 Rank Expansion

498 *Proof of Proposition 1.* Write the Tucker tensor entrywise as

$$\mathcal{S}_{i_1, \dots, i_N} = \sum_{r_1=1}^{R_1} \cdots \sum_{r_N=1}^{R_N} \mathcal{X}_{r_1, \dots, r_N} \prod_{n=1}^N A_{i_n, r_n}^{(n)}. \quad (26)$$

499 Taking an element-wise p -th power gives

$$(\mathcal{S}_{i_1, \dots, i_N})^p = \prod_{q=1}^p \left(\sum_{r_1^{(q)}=1}^{R_1} \cdots \sum_{r_N^{(q)}=1}^{R_N} \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \prod_{n=1}^N A_{i_n, r_n^{(q)}}^{(n)} \right) \quad (27)$$

$$= \sum_{\gamma_1 \in [R_1]^p} \cdots \sum_{\gamma_N \in [R_N]^p} \left(\prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}} \right) \prod_{n=1}^N \left(\prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)} \right), \quad (28)$$

500 where $\gamma_n = (r_n^{(1)}, \dots, r_n^{(p)})$. For each mode n , let $\alpha_n \in \mathbb{N}^{R_n}$ record the counts of the entries in γ_n ,
 501 so $|\alpha_n| = p$. Then

$$\prod_{q=1}^p A_{i_n, r_n^{(q)}}^{(n)} = \prod_{r=1}^{R_n} \left(A_{i_n, r}^{(n)} \right)^{\alpha_{n,r}} = A_{i_n, \alpha_n}^{(n)\langle p \rangle}. \quad (29)$$

502 Group all ordered tuples $(\gamma_1, \dots, \gamma_N)$ that produce the same multi-indices $(\alpha_1, \dots, \alpha_N)$, and define
 503 the aggregated core

$$\mathcal{X}_{\alpha_1, \dots, \alpha_N}^{(p)} = \sum_{\substack{\gamma_1, \dots, \gamma_N: \\ \text{count}(\gamma_n) = \alpha_n \ \forall n}} \prod_{q=1}^p \mathcal{X}_{r_1^{(q)}, \dots, r_N^{(q)}}. \quad (30)$$

504 The preceding display is exactly the Tucker reconstruction

$$\mathcal{S}^{\odot p} = \llbracket \mathcal{X}^{(p)}; \mathbf{A}^{(1)\langle p \rangle}, \dots, \mathbf{A}^{(N)\langle p \rangle} \rrbracket. \quad (31)$$

505 The n -th Veronese factor has $\binom{R_n+p-1}{p}$ columns, giving the claimed Tucker rank bound. $\square$

506 B.3 Separation

507 **Lemma 2** (Full-rank Veronese evaluations). *Let $D = \binom{R+p-1}{p}$. There exists a matrix $\mathbf{A} \in \mathbb{R}^{D \times R}$*
 508 *whose Veronese feature matrix $\mathbf{A}^{\langle p \rangle} \in \mathbb{R}^{D \times D}$ has rank D . Consequently, for $I \geq D$, a generic*
 509 *matrix $\mathbf{A} \in \mathbb{R}^{I \times R}$ has $\text{rank}(\mathbf{A}^{\langle p \rangle}) = D$.*

510 *Proof.* The homogeneous degree- p monomials in R variables are linearly independent polynomials.
 511 Hence there exist evaluation points $x_1, \dots, x_D \in \mathbb{R}^R$ such that the matrix $(x_i^\alpha)_{i,\alpha}$, $|\alpha| = p$, is
 512 nonsingular; otherwise every $D \times D$ evaluation determinant would vanish, which would imply a
 513 nontrivial linear dependence among the monomials. Taking these points as the rows of $\mathbf{A}$ gives the
 514 first claim.

515 For $I \geq D$, choose any D rows. The determinant of the corresponding $D \times D$ Veronese submatrix
 516 is a polynomial in the entries of $\mathbf{A}$ that is not identically zero by the first claim. Its nonvanishing is
 517 therefore a generic property, so $\mathbf{A}^{\langle p \rangle}$ has full column rank for generic $\mathbf{A}$. $\square$

518 *Proof of Theorem 1.* Let $D = \binom{R+p-1}{p}$. Choose generic factor matrices $\mathbf{A}^{(n)} = [a_1^{(n)}, \dots, a_R^{(n)}] \in$
 519 $\mathbb{R}^{I_n \times R}$, and define the CP-diagonal Tucker backbone

$$\mathcal{S} = \sum_{r=1}^R a_r^{(1)} \otimes \dots \otimes a_r^{(N)}. \quad (32)$$

520 Its mode- n unfolding is

$$\mathcal{S}_{(n)} = \mathbf{A}^{(n)} \left(\mathbf{A}^{(N)} \odot \dots \odot \mathbf{A}^{(n+1)} \odot \mathbf{A}^{(n-1)} \odot \dots \odot \mathbf{A}^{(1)} \right)^\top. \quad (33)$$

521 For generic $\mathbf{A}^{(n)}$, $\mathbf{A}^{(n)}$ has rank R , and the Khatri-Rao product on the right has rank R . Thus
 522 $\text{rank}(\mathcal{S}_{(n)}) = R$ for every n , so $\mathcal{S}$ has exact Tucker rank $(R, \dots, R)$.

523 Set $\beta_p = 1$ and $\beta_q = 0$ for $q \neq p$. Then $\mathcal{Y} = \mathcal{S}^{\odot p}$ is a valid degree- p NTD-PL tensor. Entrywise,

$$\mathcal{Y}_{i_1, \dots, i_N} = \left(\sum_{r=1}^R \prod_{n=1}^N A_{i_n, r}^{(n)} \right)^p \quad (34)$$

$$= \sum_{|\alpha|=p} c_\alpha \prod_{n=1}^N \prod_{r=1}^R \left(A_{i_n, r}^{(n)} \right)^{\alpha_r}, \quad (35)$$

524 where $c_\alpha = p! / (\alpha_1! \dots \alpha_R!) > 0$. Let $\mathbf{B}^{(n)} = \mathbf{A}^{(n)\langle p \rangle} \in \mathbb{R}^{I_n \times D}$. By Lemma 2, each $\mathbf{B}^{(n)}$ has full
 525 column rank D for generic $\mathbf{A}^{(n)}$. The mode- n unfolding is

$$\mathcal{Y}_{(n)} = \mathbf{B}^{(n)} \text{diag}(c) \left(\mathbf{B}^{(N)} \odot \dots \odot \mathbf{B}^{(n+1)} \odot \mathbf{B}^{(n-1)} \odot \dots \odot \mathbf{B}^{(1)} \right)^\top. \quad (36)$$

526 The diagonal matrix $\text{diag}(c)$ is nonsingular. Since a Khatri-Rao product of full-column-rank matrices
 527 is full column rank, the right factor has rank D . Therefore $\text{rank}(\mathcal{Y}_{(n)}) = D$. This holds for every
 528 mode n , and the exact Tucker rank of $\mathcal{Y}$ is $(D, \dots, D)$. $\square$

529 B.4 Masked Generalization

530 **Theorem 4** (Masked-entry uniform convergence). *Let $\Theta \subset \mathbb{R}^d$ be a bounded NTD-PL parameter set*
 531 *with*

$$d = \prod_{n=1}^N R_n + \sum_{n=1}^N I_n R_n + (P + 1).$$

532 For $\theta \in \Theta$, define the full-entry risk

$$\mathcal{L}(\theta) = \frac{1}{M} \sum_{\mathbf{i}} \ell_{\mathbf{i}}(\theta), \quad M = \prod_{n=1}^N I_n,$$

533 and the masked empirical risk

$$\widehat{\mathcal{L}}_m(\theta) = \frac{1}{m} \sum_{t=1}^m \ell_{\mathbf{i}_t}(\theta),$$

534 where $\mathbf{i}_t$ are sampled uniformly from tensor entries. Assume $0 \leq \ell_{\mathbf{i}}(\theta) \leq B_\ell$ and $\ell_{\mathbf{i}}(\theta)$ is G -Lipschitz
 535 in θ , uniformly over $\mathbf{i}$. If Θ is contained in a Euclidean ball of radius B_Θ , then with probability at
 536 least $1 - \delta$,

$$\sup_{\theta \in \Theta} \left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}$$

537 for every $0 < \varepsilon \leq B_\Theta$.

538 *Proof of Theorem 4.* Fix $0 < \varepsilon \leq B_\Theta$, and let $\mathcal{N}_\varepsilon$ be an ε -net of Θ in Euclidean norm. Since Θ lies
 539 in a d -dimensional ball of radius B_Θ , it admits a net with

$$|\mathcal{N}_\varepsilon| \leq \left(\frac{3B_\Theta}{\varepsilon} \right)^d. \quad (37)$$

For any fixed $\theta_0 \in \mathcal{N}_\varepsilon$, the random variables $\ell_{it}(\theta_0)$ are independent and bounded in $[0, B_\ell]$. Hoeffding's inequality gives

$$\Pr \left(\left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \geq u \right) \leq 2 \exp \left(-\frac{2mu^2}{B_\ell^2} \right). \quad (38)$$

Taking a union bound over $\mathcal{N}_\varepsilon$, with probability at least $1 - \delta$,

$$\max_{\theta_0 \in \mathcal{N}_\varepsilon} \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \leq B_\ell \sqrt{\frac{\log(2|\mathcal{N}_\varepsilon|/\delta)}{2m}}. \quad (39)$$

Substituting the net-size bound yields the stated stochastic term.

Now take any $\theta \in \Theta$, and choose $\theta_0 \in \mathcal{N}_\varepsilon$ with $\|\theta - \theta_0\|_2 \leq \varepsilon$. By the uniform Lipschitz assumption,

$$|\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| \leq G\varepsilon, \quad |\widehat{\mathcal{L}}_m(\theta) - \widehat{\mathcal{L}}_m(\theta_0)| \leq G\varepsilon. \quad (40)$$

Therefore

$$\left| \mathcal{L}(\theta) - \widehat{\mathcal{L}}_m(\theta) \right| \leq |\mathcal{L}(\theta) - \mathcal{L}(\theta_0)| + \left| \mathcal{L}(\theta_0) - \widehat{\mathcal{L}}_m(\theta_0) \right| \quad (41)$$

$$+ |\widehat{\mathcal{L}}_m(\theta_0) - \widehat{\mathcal{L}}_m(\theta)| \quad (42)$$

$$\leq 2G\varepsilon + B_\ell \sqrt{\frac{d \log(3B_\Theta/\varepsilon) + \log(2/\delta)}{2m}}. \quad (43)$$

Taking the supremum over $\theta \in \Theta$ completes the proof. $\square$

Remark 1. The Lipschitz and bounded-loss assumptions are standard for compact parametric classes. For NTD-PL, they follow when the observed values, core, factor matrices, and polynomial coefficients are bounded and the polynomial degree P is finite. The resulting Lipschitz constant may depend polynomially on these bounds and on P , but the covering dimension remains the number of tied NTD-PL parameters rather than the number of parameters in an untied high-order Tucker expansion.

C Additional Experiments

C.1 Controlled Synthetic Mechanism Checks

The controlled experiments in the main text are designed to isolate the role of the scalar link. Table 4 first checks a negative control: when the source is linear Tucker, NTD-PL does not improve by inventing nonlinear terms. The only exception is an affine offset, where the constant link term is the intended correction.

Table 4: Linear consistency on synthetic Tucker tensors. In the linear negative control, effective $p > 1$ contributions remain small except for the intended affine offset case.

Method	bias = 0		bias = 0.5	
	RMSE↓	$p > 1$	RMSE↓	$p > 1$
Tucker	0.0114	—	0.0647	—
NTD-PL, $\beta_0 = 0$	0.0116	4.19%	0.0522	39.17%
NTD-PL, β_0 free	0.0117	3.19%	0.0118	2.98%

For the nonlinear controlled tensors, the nonlinearity is introduced as an orthogonal residual rather than by directly setting $Y = g(S^*)$. Let S^* be the linear low-rank Tucker backbone and let g be one of the candidate scalar responses. We form

$$R^* = g(S^*) - \frac{\langle g(S^*), S^* \rangle}{\|S^*\|_F^2} S^*, \quad (44)$$

and, when $\|R^*\|_F > 0$, normalize it as

$$\tilde{R}^* = \frac{\|S^*\|_F}{\|R^*\|_F} R^*. \quad (45)$$

563 The observed tensor is then

$$Y_\alpha = \sqrt{1-\alpha} S^* + \sqrt{\alpha} \tilde{R}^*, \quad 0 \leq \alpha \leq 1. \quad (46)$$

564 By construction, $\langle S^*, \tilde{R}^* \rangle = 0$ and $\|\tilde{R}^*\|_F = \|S^*\|_F$, so α is exactly the energy fraction assigned to
 565 the nonlinear residual. This keeps the total energy fixed while moving the observation away from the
 566 linear Tucker backbone. The candidate residual families are $g(s) = s^2 + s^3$, $g(s) = (\exp(\kappa s) - 1)/\kappa$,
 567 and $g(s) = \tanh(\kappa s)$, with κ set from the input range.

568 The main text reports the nonlinear alpha sweep and degree sweep. Figure 6 shows that degree contin-
 569 uation activates higher powers with systematic loss drops at relevant degrees. Table 5 complements
 570 those figures with a small but consistent term-mass concentration effect: the dominant contribution
 571 usually lands on the degree that matches the injected response, while smooth odd links spread mass
 572 across nearby odd orders.

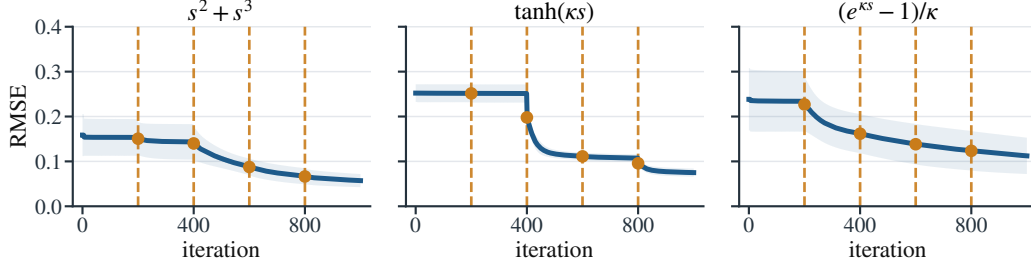


Figure 6: NTD-PL training curves for controlled nonlinear mechanism checks at $\alpha = 0.25$ and $P = 5$. Dashed vertical lines mark degree activations, which produce systematic drops when the activated degree matches the response structure.

Table 5: Effective NTD-PL term contributions under moderate nonlinearity, reported as percentages over ten seeds. Polynomial targets concentrate mass at matching nonlinear degrees, while odd smooth links place most nonlinear mass on odd powers.

Link	Effective contribution (%)						
	$p = 0$	$p = 1$	$p = 2$	$p = 3$	$p = 4$	$p = 5$	$p = 6$
$s^2 + s^3$	0.1	24.1	9.8	38.1	7.3	11.5	9.1
$(e^{\kappa s} - 1)/\kappa$	0.3	39.8	16.8	13.2	10.1	7.7	12.1
$\tanh(\kappa s)$	0.2	37.9	1.3	30.9	5.4	17.3	7.0

573 C.2 CAVE Completion Significance and Robustness

574 The main text reports the full-resolution random-missing CAVE completion table at rank $(33, 33, 4)$.
 575 We do not repeat that table here; instead, Table 6 gives the paired sign and Wilcoxon tests for the
 576 same protocol.

Table 6: Paired significance tests for full-resolution CAVE completion at rank $(33, 33, 4)$. Metrics are computed only on missing entries.

ρ	RMSE				SAM			
	W/L	Med. gain	Sign p	Wilcoxon p	W/L	Med. gain	Sign p	Wilcoxon p
0.1	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	14/1	2.9742	9.77×10^{-4}	8.54×10^{-4}
0.3	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	13/2	1.9194	0.007	0.004
0.5	14/1	0.0026	9.77×10^{-4}	1.22×10^{-4}	13/2	1.6566	0.007	0.012
0.7	14/1	0.0027	9.77×10^{-4}	1.22×10^{-4}	12/3	1.3394	0.035	0.048

577 C.3 Link Refresh Residual Diagnostic

578 The diagnostic used in the main text reuses the Link Refresh step from Section 5 as a one-round
 579 residual probe. Let $\hat{\mathcal{Y}}_T$ denote the Tucker output and $\mathcal{Y}$ the target tensor. Starting from this fitted

580 Tucker latent tensor, we update only the scalar polynomial coefficients by fitting

$$h_\gamma(x) = \sum_{p=0}^4 \gamma_p x^p$$

581 by ridge regression on pairs $(\hat{Y}_{T,i}, Y_i)$. In full reconstruction, all entries are used for this fit. In
 582 completion, the fit uses only observed entries and is then evaluated on held-out entries. The fitted
 583 map is applied entrywise to $\hat{Y}_T$. We summarize this cheap refresh by the Link Refresh diagnostic

$$D_{\text{link}} = 10 \log_{10} \left(\frac{\|\mathcal{Y}_{\text{eval}} - \hat{\mathcal{Y}}_T\|_F^2}{\|\mathcal{Y}_{\text{eval}} - \hat{\mathcal{Y}}_{\text{link}}\|_F^2} \right),$$

584 which measures, on a log scale, how much Tucker residual energy is removed on the task-specific
 585 evaluation entries. In other words, it measures how much shared scalar-link yield is already present
 586 before the full NTD-PL alternating solver starts reshaping the Tucker parameters.

587 C.4 CAVE Completion Scene Diagnostics

588 Figure 7 supplements the main CAVE diagnostic table with the scene-level scatter plots used to
 589 visualize the link yield across missing rates. The plots show the same diagnostic relationship between
 590 D_{link} and held-out RMSE* gain, now at the scene level.

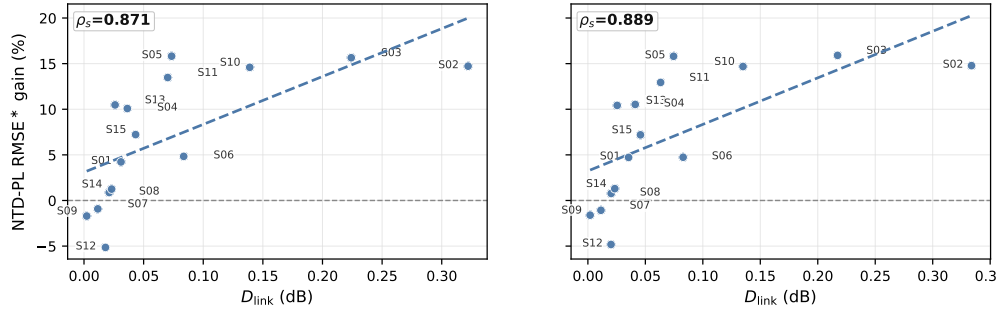


Figure 7: Scene-level supplement for the CAVE completion diagnostic table. The left and right panels use missing rates 0.3 and 0.5, respectively; D_{link} from frozen-backbone Link Refresh predicts larger held-out RMSE* gains.

591 C.5 Non-HSI Link Refresh Diagnostics

592 The main diagnostic figure uses per-unit non-HSI experiments rather than collection-level tensors.
 593 CBSD68 [Martin et al., 2001] uses per-image units resized to $96 \times 96 \times 3$ with rank $(32, 32, 3)$.
 594 CIFAR-10 [Krizhevsky, 2009] uses 1000 class-balanced test images of shape $32 \times 32 \times 3$ with rank
 595 $(20, 20, 3)$. COIL-100 [Nene et al., 1996] uses one object at a time as a $36 \times 48 \times 48 \times 3$ view tensor
 596 with rank $(8, 12, 12, 2)$. smallNORB [LeCun et al., 2004] uses one object instance at a time as an
 597 $18 \times 64 \times 64 \times 2$ stereo view tensor with rank $(12, 24, 24, 2)$. KTH-Action [Schuldt et al., 2004]
 598 and UCF101 [Soomro et al., 2012] provide the action-video tensor units used in the main diagnostic
 599 scatter. All units are max-normalized independently, and diagnostic labels are assigned only from
 600 D_{link} , not from NTD-PL gain.

Table 7: Dataset-level statistics behind the non-HSI diagnostic scatter in Figure 4. The gain columns report mean NTD-PL RMSE improvement over independently fitted matched-rank Tucker within D_{link} terciles.

Domain	Dataset	Units	Med. D_{link}	ρ_s	Low	Mid	High
Natural images	CBSD68	68	0.091	0.85	4.17%	8.40%	14.60%
Natural images	CIFAR-10	1000	0.008	0.93	-0.01%	2.59%	7.85%
Object-view	COIL-100	100	0.208	0.93	2.77%	5.46%	9.14%
Object-view	smallNORB	25	0.264	0.95	6.92%	9.91%	14.16%
Action video	KTH-Action	240	0.107	0.88	2.74%	4.27%	10.87%
Action video	UCF101	240	0.065	0.89	1.44%	2.97%	5.89%

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